

Hi

In this segment, we will have a look at policy iteration algorithm.

First, we import the libraries. The environment is a 4x4 gridworld with two terminal states, marked by "T".

Then, we create an instance of this gridworld.

In order to behave optimally in this environment, or any environment, we need an optimal policy. Recall that policy evaluation algorithm computes value function, but it doesn't provide the optimal policy.

Consider this: Given a policy, if there was some process to get even a slightly better policy, we would repeat such a process iteratively to get better and better policies, all the way to optimal policy.

It turns out that such a process exists. A policy which is greedy w.r.t. a value function V_{π} turns out to be better than the policy π .

Also note that when improved policy becomes the same as old policy, optimality is reached.

Armed with this knowledge, let's look at the policy iteration algorithm.

We start with arbitrary value function and policy.

Policy evaluation phase is the same as before.

In policy improvement phase, we build an improved policy as follows:

For each state, we assign the action which results in the maximum return according to the value function. Since we are only selecting the best action and ignoring the other actions, we say that we are acting greedily w.r.t. the value function.

If the new policy is same as old policy, we stop. Otherwise, we repeat from the policy evaluation phase.

As for the code, policy evaluation code is same as before.

For policy improvement, the `one_step_lookahead` method calculates the returns of all the actions in a state.

After that, we start with a random policy. We evaluate that policy and construct a new improved policy with the help of `one_step_lookahead`. When optimality is reached, we stop and return the policy and the corresponding value function.

Now, we run the algorithm and get the policy. Next, we make sure that it is close to the expected values.

Then, we plot both the calculated and expected value functions.

Now, we see the optimal policy in action.

It reaches the goal.

To summarize, we used the policy evaluation algorithm in addition to an improvement step to construct the policy iteration algorithm. Later, we will look at a value-function-based method to obtain the optimal policy. Thank you.